Establishing Phase Definitions for Jump and Drop Landings and an Exploratory Assessment of Performance-Related Metrics to Monitor During Testing

Article *in* The Journal of Strength and Conditioning Research · December 2023

DOI: 10.1519/JSC.0000000000004700

John R Harry
Texas Tech University

the Journal of Strength and Conditioning Research™

Establishing Phase Definitions for Jump and Drop Landings and an Exploratory Assessment of Performance-Related Metrics to Monitor During Testing

John R. Harry, Anton Simms, and Mia Hite

Human Performance & Biomechanics Laboratory, Department of Kinesiology & Sport Management, Texas Tech University, Lubbock, TX

Abstract

Harry, JR, Simms, A, and Hite, M. Establishing phase definitions for jump and drop landings and an exploratory assessment of performance-related metrics to monitor during testing. *J Strength Cond Res* XX(X): 000–000, 2023—Landing is a common task performed in research, physical training, and competitive sporting scenarios. However, few have attempted to explore landing mechanics beyond its hypothesized link to injury potential, which ignores the key performance qualities that contribute to performance, or how quickly a landing can be completed. This is because a lack of (a) established landing phases from which important performance and injury risk metrics can be extracted and (b) metrics known to have a correlation with performance. As such, this article had 2 purposes. The first purpose was to use force platform data to identify easily extractable and understandable landing phases that contain metrics linked to both task performance and overuse injury potential. The second purpose was to explore performance-related metrics to monitor during testing. Both purposes were pursued using force platform data for the landing portion of 270 jump-landing trials performed by a sample of 14 NCAA Division 1 men's basketball players (1.98 ± 0.07 m; 94.73 ± 8.01 kg). The proposed phases can separate both jump-landing and drop-landing tasks into loading, attenuation, and control phases that consider the way vertical ground reaction force (GRF) is purposefully manipulated by the athlete, which current phase definitions fail to consider. For the second purpose, Pearson's correlation coefficients, the corresponding statistical probabilities ($\alpha = 0.05$), and a standardized strength interpretation scale for correlation coefficients ($0 <$ trivial $\leq 0.1 <$ small $\leq 0.3 <$ moderate $\leq 0.5 <$ large $\leq 0.7 <$ very large) were used for both the group average (i.e., all individual averages pooled together) and individual data (i.e., each individual's trials pooled together). Results revealed that landing time, attenuation phase time, average vertical GRF during landing, average vertical GRF during the attenuation phase, average vertical GRF during the control phase, vertical GRF attenuation rate, and the amortization GRF (i.e., GRF at zero velocity) significantly correlated with landing performance, defined as the ratio of landing height and landing time ($R \geq \pm 0.58$; $p < 0.05$), such that favorable changes in those metrics were associated with better performance. This work provides practitioners with 2 abilities. First, practitioners currently assess jump capacity using jump-landing tests (e.g., countermovement jump) with an analysis strategy that makes use of landing data. Second, this work provides preliminary data to guide others when initially exploring landing test results before identifying metrics chosen for their own analysis.

Key Words: ground reaction force, jumping, force platform

Introduction

Landing is a task inherent to most ground-based sports and physical activities that involve jumping actions (11), such as basketball, soccer, and volleyball, to name only a few. Landing must be completed following a jumping action, which is important because the countermovement jump is arguably the most popular task performed in athlete testing scenarios (35). As such, practitioners can easily include landings when testing an athlete's physical capacity. One reason to include landings to assess athletes' capacity relates to their broad potential for musculoskeletal injury. Specific risk for certain tissues, such as anterior cruciate ligament injury or other structural tissues, can only be revealed by force platforms with combining motion capture (27,40) that might not be accessible to practitioners. However, there continues to be a growing availability and use of research-grade force platform technology (35). This availability provides practitioners with the means to extract ground reaction force (GRF) metrics often linked to musculoskeletal injury potential (6,9–11,33,36–38,44,45,50). Although GRF metrics do not have a clear causal relationship with musculoskeletal injury occurrences (15), they can be explored to design or modify athlete training interventions.

In our view, the main reason to include landings when testing an athlete's physical capacity is the importance of completing a landing quickly during competition, which is why landing "performance" is defined as the time required to terminate downward motion upon ground contact (15,16). Athletes must terminate downward motion as rapidly as possible so that a secondary action can be performed if needed. Repeated jumps (i.e., jump landing followed

Address correspondence to John R. Harry, john.harry@ttu.edu.

Journal of Strength and Conditioning Research 00(00)/1–10

immediately by a second jump landing), for example, are common to sports like basketball and soccer among others (7,17). If the athletes involved in such tasks possessed the capacity to complete the landing portion more quickly than their opponent, they would achieve greater potential for success during the secondary movement. For example, consider 2 athletes who have the same ground contact time (i.e., time between the start landing and the subsequent jump) and height achieved during the subsequent jump. The athlete who terminates their downward motion the quickest, meaning they completed the landing faster, employed a strategy facilitating a greater portion of the ground contact time to propulsive force application and ultimately a greater jump height. Given the fact that soccer players perform jump landings ~11 times per match from corner kicks alone (47) and basketball players can perform jump landings 1.5 times per minute during a competitive game (8), this is a valuable capacity for practitioners to assess and target for improvement in their athletes.

The need to perform a landing safely and quickly seems obvious. However, few attempts have been made by researchers to reveal how athletes strategize GRF accommodation to minimize musculoskeletal injury potential, rapidly terminate their downward motion, or both. Relative to musculoskeletal injury potential, vertical GRF during landing can be used alongside newtonian predictions to classify athletes' landing strategies according to whether they experience more than, less than, or equal to the predicted amount of loading (29,41). These classifications provide practitioners with the awareness of inappropriate GRF accommodation patterns that might require intervention to help athletes better organize the way they accommodate landing impact. For rapid landing performance, individuals with better landing performances were observed to have greater rates of force attenuation, which is the rate of change of vertical GRF between the peak vertical GRF and the end downward motion (15). The rate of force attenuation is a "yank" metric, similar to the popular rate of force development metric assessed by many when testing vertical jump capacity (1,17,31). Individuals with better landing performances were also observed to use greater knee and ankle contributions to energy absorption than those with poorer landing performances during the time between ground contact and the peak vertical GRF magnitude (loading) and the time between the peak vertical GRF magnitude and the end of downward motion (attenuation). For the strength and conditioning practitioner, assessing landing strategies using these approaches can enhance the ability to make, and the effectiveness of, actionable decisions.

Although the studies summarized in the previous paragraph (15,29,41) provide some insight into how athletes strategize landing GRF accommodation and performance, each used either the entire landing period or an overly general approach to deconstruct the landing into phases. Mechanically, the objective during landing is to accommodate the impact forces such that the task objective is not compromised. At face value, this means the impact force experienced, which rapidly increases immediately after ground contact, must be decreased. This logic framed the phase deconstruction presented previously (15). However, the vertical GRF magnitudes do not simply increase and decrease during landing, as they instead increase, decrease, and increase again (Figure 1). This means that potentially important vertical GRF manipulation strategies and within-landing task objectives can be ignored if only loading and attenuation phases are used as defined previously. A clear set of phases that provide the ability to assess performance and performance-related strategies would be useful for the strength and conditioning practitioners currently taking advantage of force platform technology to perform countermovement jump tests. A key feature of any set of landing phases is that the phases should be specific to the GRF time profile. This will help to ensure that researchers and practitioners can examine landing performance using similar and sound approaches without the need to also have access to motion capture technology. This will provide a means to compare research results with test results in the field.

The purpose of this study was 2-fold. First, we sought to present easily extractable and understandable phases of landing that provide measures linked to task performance without compromising the ability to extract metrics linked to overuse injury potential. Second, we explored performance-related metrics that could be monitored during testing. Both purposes were achieved using data obtained from a sample of NCAA Division 1 men's basketball players. Similar studies have been published for the propulsive portion of countermovement vertical jump (18,35), and those have helped to create consistency and ecological validity among studies or practical tests. This study seeks to do the same for landing, using GRF data obtained from a collegiate athlete population.

Methods

Experimental Approach to the Problem

To present easily extractable and understandable phases of landing, we obtained GRF data during biweekly testing sessions over a 6-month period. From those data, we discuss necessary data collection procedures and the required characteristics of the GRF-time curve. We also thoroughly define and describe the phases of landing from functional and mechanical perspectives. To explore performance-related metrics that could be monitored during testing, we used the same data set to obtain landing performance metrics, strategy metrics commonly used in the landing literature, and additional metrics from within our proposed phases. Pearson's product-moment correlation coefficients were used to determine the relationships among metrics, focusing on both the strength and direction of associations. Ultimately, we sought to provide preliminary data to guide practitioners when initially exploring landing test results before identifying metrics chosen for their own analysis.

Subjects

We elected to achieve both purposes of this article using GRF data collected from a convenience sample of 14 NCAA Division 1 men's basketball players (average height: 1.98 ± 0.07 m; average body mass: 94.73 ± 8.01 kg). All were healthy and active members of the basketball program at the same university throughout the duration of the data collection period. As some players were freshmen or transfers while others were returners, it was not possible to obtain reliable training history information. Nonetheless, the sample is like many NCAA basketball programs. All provided written informed consent. Approval was obtained from the Texas Tech University Institutional Review Board at the site of data collection in accordance with the Declaration of Helsinki.

Procedures

To obtain a large data set, biweekly testing sessions were conducted for a 6-month period starting in August (preseason) and

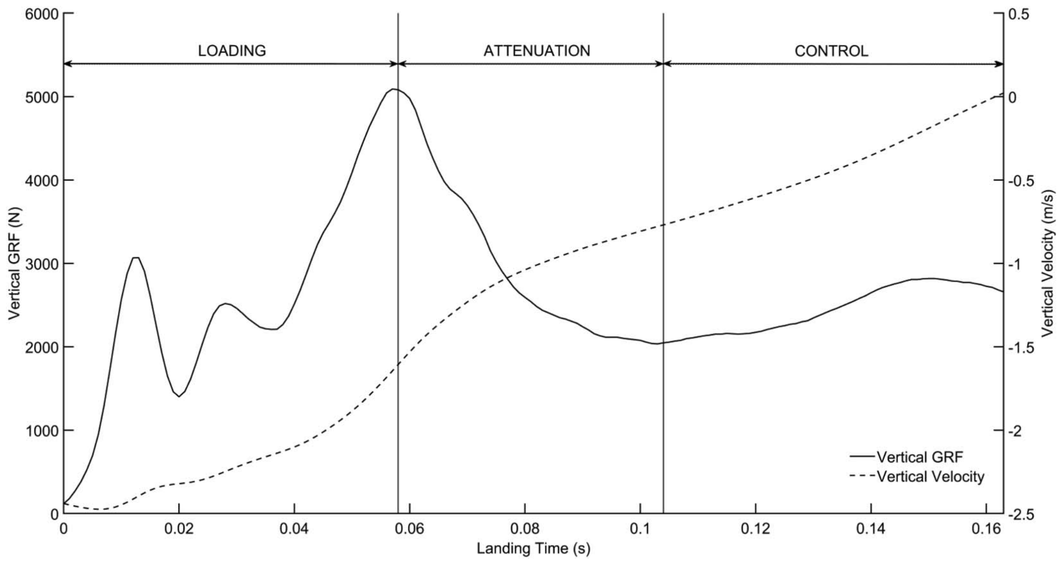

Figure 1. Exemplar vertical GRF and COM velocity curves during landing with key phases. GRF = ground reaction force; COM = center of mass.

continuing through to the following February (midseason), with in-season tests conducted ~48 hours before or after a competitive game. During each session, subjects completed a standardized warm-up under the direction of their team's head strength and conditioning coach. Because of the individual needs of each athlete or the previous day's training volume or intensity, the warm-up could vary from test to test to adequately prepare the players for testing and subsequent physical training. Players then performed a vertical jump-landing test, in which a maximum-effort countermovement rebound jump was performed (i.e., countermovement jump + reactive jump + landing), with only the final landing used here. Each jump began with players standing upright as motionless as possible. They were instructed to perform the jump with their preferred arm swing and countermovement depth strategies seeking to achieve maximum vertical height as quickly as possible. Upon landing, subjects were instructed to terminate their downward motion as quickly as possible again using their preferred landing depth and arm motion or positioning. No verbal encouragement or performance-related feedback was provided. Three trials were performed during each test, with up to 30 seconds of rest provided between the trials as needed. During each trial, vertical GRF data were collected using a 3-dimensional force platform (Accupower; Advanced Mechanical Technology, Inc. [AMTI], Watertown, MA) at a sampling rate of 1,000 Hz and interfaced to a personal computer running AMTI NetForce acquisition software. Trials were discarded and repeated if a subject could not return to a standing motionless position after the landing or was judged (by researchers or by their own perception) to have performed the trial at a submaximal intensity. Across the sample, the average and standard deviation for the number of trials included was 19 ± 5 trials, with a range of 9–24 trials.

PART 1: Establishing Landing Phase Definitions

Necessary Characteristics and Motion Qualities Obtained from Recorded Force-Time Curves. Research involving landings usually have subjects perform the task from a raised platform, described as a "drop landing" (9,10,50). Although it is our opinion that landings should be performed following a vertical jump because of dissimilar mechanics between drop landings and jump landings (22,23), the phase deconstruction method can be defined in the same way for both tasks. As mentioned, the usability of any phase deconstruction method intended for strength and conditioning practitioners is dependent on the ability to identify the phases using only force platform data (18,35), and we have taken this approach here.

To ensure that the landing can be deconstructed into phases to achieve the first purpose of this article, certain features of the recorded GRF data must be present. There must be ~0.5–1.0 seconds of time at the end of the trial where the performer is standing upright and stationary (13). Mean vertical force can be calculated from all of the vertical GRF data points within the 0.5 seconds or 1.0 seconds time period to obtain body mass (19). Although jump landings can use the 0.5- to 1.0-second time period before the jumping action for these calculations, it is important to maintain usability and comparability for drop landings and jump landings because drop landings will not have any GRF readings before ground contact (13). Body mass must also be calculated by dividing the body mass value by the absolute value of gravitational acceleration (mass = BW/9.81). Vertical center of mass (COM) acceleration is then calculated at each data point using Newton's law of acceleration (a = $\sum$Force/mass), with vertical COM velocity obtained as the cumulative time integral of the acceleration (31).

Isolating the Landing Action. A subjective vertical GRF threshold (e.g., 10 N or 20 N) has been used to define ground contact (i.e., the start of landing) in the majority of landing-specific research and testing protocols (15–17,22–24). However, recent research has begun to objectively define ground contact on a trial-to-trial basis using the average vertical GRF plus 5 standard deviations calculated during an exemplar time window during the flight phase period when the jumper is off the force platform (19,32), and we recommend using this kind of approach. For these data, we obtained the flight phase threshold from the time starting and ending 150 and 50 milliseconds, respectively, before a surrogate ground contact event (vertical GRF >50). By ignoring the first 50 ms after the surrogate takeoff event and the vertical GRF data 150 milliseconds after that event, 100 milliseconds of flight phase data from the "middle" of flight was used to calculate the ground contact event threshold. This was our way to isolate the vertical GRF and remove data points that were clearly not representative of the force platform's measurements without any load atop it.

The impulse-momentum relationship is the foundation for our definition for the end of the landing action (33). This is because the athlete's velocity and momentum developed before ground contact is decreased and ultimately terminated through lower-extremity joint work (24) and external force attenuation. The amount of net impulse applied is equal to the change of momentum, meaning the end of landing occurs when vertical COM velocity (and therefore momentum) crosses zero after ground contact. This event defines the change of direction from downward to upward. Any motion occurring beyond this time point is no longer performed with the intention of attenuating the impact or terminating downward motion, such as the aforementioned landing followed by a sprint. Instead, the performer's goal after landing and terminating downward motion is to maintain balance and return to and maintain a standing posture (33). This time period is often combined with landing, as defined here, and used to extract a metric known as the time to stabilization (49). It is our opinion that the time to stabilization be reserved for clinical or rehabilitation objectives and should not be considered part of landing, especially in healthy athlete populations. With ground contact and the end of landing established, the 3 key phases of landing can be extracted, which we refer to as *loading, attenuation, and control.*

The Loading Phase. Loading describes the impact accommodation time associated with the performer's initial collision with the ground to terminate their momentum created by gravity during the downward part of flight before ground contact. Therefore, it is defined as the time between ground contact and the peak vertical GRF magnitude, as shown in Figure 1. Despite a lack of a formal definition for the phase, the loading phase has historically been of interest for 2 primary reasons. First, exposing the body to large-magnitude impact forces are believed to increase the potential for injury, such as fractures or structural tissue degradation (39). During a typical loading, the vertical GRF values are quite large, with the first peak magnitude (sometimes called "F1"), which represents the force attenuation by the forefoot only, typically reaching ~3 times body mass (11). The global peak (sometimes called "F2"), which represents force attenuation by the rearfoot and lower limbs, typically reaches ~7 times body mass during a typical jump landing (15). Interestingly, those magnitudes, particularly the largest peak (i.e., F2), are ~11% greater during drop landings and jump landings from equal heights (23). Second, these peak vertical GRF magnitudes are experienced very soon after ground contact, usually between 0.05 and 0.07 seconds (10,11,17,23). The change of vertical GRF between ground contact and the peak magnitude is divided by the time to reach the peak GRF to obtain a metric referred to as the loading rate, and these metrics are commonplace among studies seeking to estimate musculoskeletal injury potential (15,16).

The impulse-momentum relationship can again be referred to when seeking to explore loading-related performance qualities. In particular, the magnitude of net vertical GRF impulse throughout loading is the first of 3 impulse values (i.e., one per phase) used to create a sum that reflects the effort applied to complete the landing. However, to adequately interpret loading impulse, the average vertical GRF and loading time data should be considered in parallel to determine how an athlete strategized their effort during this phase. Based on recent results (15), performers who complete landings more quickly (i.e., higher-performance output) tend to employ a strategy characterized by decreasing the amount of impulse during loading. This means that the overall impulse applied throughout the entire landing is organized such that a smaller portion is applied during loading. This means those with poorer landing performance, and therefore greater loading phase impulse, likely experience increased average vertical GRF and similar or reduced loading times. This would indicate greater GRF-specific musculoskeletal injury potential (e.g., stress fracture) because there would be greater impact force and loading rate values. Furthermore, lesser loading impulse by way of similar or reduced vertical GRF and similar loading times would indicate reduced musculoskeletal injury potential because of smaller impact force and loading rate values. Ultimately, strategies characterized by lesser loading impulse implies more effort application during the postloading phases.

It is important for practitioners to note that the performer has limited control over the vertical GRF, time to the peak vertical GRF, and therefore the loading impulse, after contact with the ground. This is because executing a voluntary movement requires more time than what is available during loading (48). For overuse injury risk reduction interventions, any technique-related changes sought within the loading phase must be stimulated through preparatory impact strategies during the preimpact period (22). One simple strategy is increasing plantarflexion before ground contact, which can significantly reduce the peak vertical GRF magnitude and the loading rate (44). These beneficial loading changes from increasing plantarflexion at contact seem to occur even when the knee joint is rigidly positioned with less flexion at contact (46). This is one example of a preparatory strategy that positively affects overuse injury risk.

The Attenuation Phase. The attenuation phase was originally presented as the time between the end of loading (i.e., the peak vertical GRF) and the end of the landing (15). That study revealed that those who can complete a landing more quickly have more vigorous force manipulation and perform a greater portion of the work done by the knee joint and a lesser portion by the hip joint than those who land slowly. In soccer athletes, moderate-to-large differences in force manipulation have been observed among athletes of different primary positions, when that same phase deconstruction method was used (16). However, we have come to realize that defining the attenuation end point by the end of the landing is inaccurate. When inspecting the typical vertical GRF and velocity curves during landing (Figure 1), a performer's force manipulation after the start of attenuation involves a decrease in vertical GRF followed by an increase in vertical GRF. This means that force is not just attenuated between the start of loading through the end of the task. Because of these observations, we recommend redefining the attenuation phase as the time between the end of loading (i.e., the peak vertical GRF) and the local minimum vertical GRF (i.e., the first valley after the peak vertical GRF). This is because before the local minimum vertical GRF, force attenuation occurs by reducing the amount of force applied to the ground while causing upward acceleration to slow the body's downward velocity. After the local minimum vertical GRF, force development occurs by increasing the amount of force applied to the ground, but because the force magnitudes are below body mass, the performer is still moving downward. Like the loading phase, the average vertical GRF and the time of the phase can be important metrics because of previous findings, indicating that the rate of vertical GRF attenuation distinguishes good from poor performers during landing (15). This also means both the ratio of those 2 metrics (i.e., the rate of vertical GRF attenuation) and the product of the 2 metrics (i.e., the net attenuation impulse) are important metrics that reveal phase-related change. However, like the loading phase, the component parts of the attenuation rate and attenuation impulse values should be explored in parallel to understand specific strategy changes.

The Control Phase. As mentioned, the attenuation phase of landing was originally defined as a longer period than we propose herein (15,16). However, the increase in vertical GRF after it reaches the local minimum describes a distinct force-related objective to increase the amount of force applied into the ground to decelerate the body's downward velocity. For example, the performer has attenuated the GRF and the local minimum vertical GRF occurs. The subsequent increase in vertical GRF relates to a transitionary period during which the performer controls their force output and determines whether the mechanical energy absorbed during their downward motion should be reused to stand (with a large portion of the energy lost as heat) or to perform another vigorous action (e.g., another jump, lateral sprint, etc). Accordingly, we have labeled this final phase of landing the *control* phase. It is important to note that, to our knowledge, this is the first formal discussion of this period within the complete landing sequence. While inspecting the data from each athlete, we noticed that a small sample of trials (8 of 270 or ~3%) did not have a control phase, and the landing was completed at the end of the attenuation phase. Practitioners are encouraged to consider how landings are instructed, as we did not provide specific instructions to the athletes, and some might have used unconventional strategies. Although motion capture data were not used here, examination of GRF-time curves from other work suggests that strategies that involve "stiff" joints do not involve a control phase (9).

Part 2: Metrics Associated With Landing Performance

The second purpose of this article was to explore performance-related metrics to monitor during testing; the sample described in part 1 of this article was again examined. The raw GRF data, obtained using the procedures described previously, were exported to MATLAB (The Mathworks, Inc., Natick, MA) for offline analysis, using a modified version of the recently published script for jump-landing analyses (14). The raw GRF data were not filtered to avoid inappropriate changes to the shape of the force-time curve (20). The calculations mentioned in part 1 for vertical COM acceleration and velocity were performed, and COM vertical position was also calculated as the cumulative time integral of the COM velocity using the trapezoidal rule. In addition, the end points discussed previously in part 1 to isolate the overall landing and the loading, attenuation, and control phases (Figure 1) were identified, and the reader is referred to Table 1 for a written summary of those time points and the definition or calculation for all metrics of interest. The metrics extracted for this study were selected through our theoretical understanding of the task in consideration with countermovement jump metrics (19) that relate to similar changes of force or related quantities.

Statistical Analyses

Average and standard deviation values for each metric of interest were calculated across all trials and sessions per subject, as our goal was not to determine adaptation but rather the association between the metrics and landing performance. As mentioned previously, across the sample, the average and standard deviation for the number of trials included was 19 ± 5 trials, with a range of 9–24 trials. Limitations related to group average analyses, especially with small samples like the one here and in most team sport settings, have been discussed and explored at length elsewhere (2–4,12,21). Accordingly, we sought to determine the relationship between landing performance and the remaining metrics of interest at the group average and individual athlete levels, similar to a recent article exploring relationships during jumping (19). Landing performance has been previously defined as the time to complete the landing (15–17). However, similar to the way countermovement jump researchers consider the height achieved relative to the time required to leave the ground (i.e., reactive strength index modified) to be a main performance metric for athletes (5,30), landing performance might best be described by the ratio of landing height and the time to complete the landing. To our knowledge, this ratio for landing has yet to be explored, but the logic behind it is sound enough with a strong basis from the jumping literature for its assessment here. As such, the ratio of landing height (i.e., downward center of mass displacement during flight) and landing time, which we call the

Table 1
Metrics of interest and associated calculations.*

Metric	Calculation or definition
Landing performance index	Landing height or landing time
Landing height	Impact velocity^2 per 19.62
Landing time	Total time between ground contact and the time vertical COM velocity crosses zero
Landing momentum	Body mass × impact velocity
Landing depth	COM vertical displacement between ground contact and end of landing
Loading time	Total time between ground contact and peak vertical GRF
Attenuation time	Total time between the peak vertical GRF and the local minimum vertical GRF
Control time	Total time between the local minimum vertical GRF and the time vertical COM velocity crosses zero
Loading rate	Average vertical GRF during loading divided by loading time
Attenuation rate	Average vertical GRF during attenuation divided by attenuation time
Control rate	Average vertical GRF during control divided by control time
Loading impulse	Time integral of net vertical GRF during loading
Attenuation impulse	Time integral of net vertical GRF during attenuation
Control impulse	Time integral of net vertical GRF during control
Average landing GRF	Average vertical GRF during the overall landing time
Average loading GRF	Average vertical GRF during the loading phase
Peak loading GRF	Peak vertical GRF during the loading phase
Average attenuation GRF	Average vertical GRF during the attenuation phase
Average control GRF	Average vertical GRF during the control phase
Amortization GRF	Vertical GRF at the end of landing (i.e., at zero vertical COM velocity)

*COM = center of mass; GRF = ground reaction force.

landing performance index (LPI), is included as the primary performance metric.

To assess the relationship between the LPI and the other metrics of interest, Pearson's product-moment correlation coefficients were calculated to report the strength and direction of association. Data normality was assessed at the group level using the Shapiro-Wilk test, and linearity was inspected using scatterplots. A 5% threshold for statistical significance ($\alpha = 0.05$) was used. Consistent with related work (19), we did not include an alpha adjustment because there is no way to localize random results (42). In addition, we elected not to bootstrap the data if normality was violated because of our combined use of group and individual assessments (4,43). We instead considered the correlations with respect to Hopkins scale (28) to identify important results, which we defined as significant correlations of very large strength ($0 <$ trivial $\leq 0.1 <$ small $\leq 0.3 <$ moderate $\leq 0.5 <$ large $\leq 0.7 <$ very large). The individual athlete data were considered a supplement to the group average data to reveal potential answers that group data could not reveal. Accordingly, metric-specific interpretations were made by generalizing the individual data when the group average results were unclear despite most athletes demonstrating similar associations.

Results

Mean and standard deviation data for each metric of interest at the group and individual levels are provided in Tables 2–5. At the group average level, the following metrics demonstrated a significant association with LPI with correlation coefficients of large or greater strength: landing time, attenuation time, average vertical GRF during landing, average vertical GRF during attenuation, average vertical GRF during control, vertical GRF attenuation rate, and the amortization GRF (i.e., GRF at zero velocity). None of the other metrics exhibited a significant association with LPI at the group average level.

At the individual athlete level, landing momentum and landing height demonstrated significant associations with LPI and correlation coefficients of large or greater strength in the same 7 athletes (50%) of the sample. Landing depth demonstrated significant associations with LPI and correlation coefficients of large or greater strength in only 4 athletes (~29%). Both landing time and attenuation time was significantly associated with LPI in the same 10 athletes (~71%) for which the correlation coefficients were of large or greater strength, whereas loading time and control time had significant and strong or greater associations in only 2 (~14%) and 4 (~28%) athletes, respectively. All 14 athletes (100%) demonstrated a significant, strong-or-greater association between average GRF during the entire landing and LPI. Average vertical GRF during loading, attenuation, and control showed significant large or greater associations with LPI in 11 (~79%), 12 (~86%), and 12 (~86%) athletes, respectively. Six of the athletes (~43%) demonstrated a significant large or greater association between the peak vertical GRF during loading and LPI, whereas 7 athletes (50%) exhibited a significant large or greater association between the loading rate and LPI. The associations between LPI and the attenuation rate and the control rate were significant and of large or greater strength in only 4 athletes (~28%) and 1 athlete (~7%), respectively. Finally, the associations between the amortization GRF (i.e., GRF at zero velocity) and LPI were significant and of large or greater strength in 9 of the athletes (~64%).

Discussion

The purpose of part 2 of this study was to explore performance-related metrics to monitor during jump-landing tests when using the phase deconstruction method described in part 1. We define landing performance herein using LPI, which is the ratio of landing height and landing time. This was done to simplify "explosive" performance assessments for practitioners currently using countermovement jump tests with the reactive strength index modified as a main performance metric. By doing so, the interpretation for each metric is the same in that an increased index is preferred, and the individual components help explain the index's change. Although landing time showed a strong and significant correlation with LPI at the group average level, landing height did not exhibit the same level of association. Specifically, the results indicate that shorter landing times likely coincide with greater LPI. This aligns with the countermovement jump literature, as greater reactive strength index modified results seem to be driven by shorter jump durations and not higher jumps (26) when arm swing is permitted. By emphasizing landing time as a key metric related to LPI, practitioners can focus on reducing the duration of the landing phase during jump-landing tasks to potentially improve "explosive" performance. However, it is important to consider other metrics as well, such as landing height and landing depth because they can provide additional insights into athletes' strategies when executing a landing.

As landing time appears vital to LPI, the phase-specific durations revealed that the period within the landing that should be the focal point of training is the attenuation phase, as it was the only phase with a significant and large-or-greater association with LPI. Single jump plyometric training is a potential intervention for this purpose based on results suggesting that jump-landing height and landing time countermovement jump increased and decrease by ~16 and ~8%, respectively, after 6 weeks of training (34). Based on the current data, landing interventions such as single plyometric training should stimulate adaptations that can decrease the time required to attenuate the vertical GRF. Interestingly, the results highlighted a significant and very strong relationship between LPI and the average vertical GRF during the attenuation phase, with greater average forces linked to greater LPI. Single jump plyometric training seems to stimulate greater average vertical GRF based on training-induced changes in the peak vertical GRF (34). This relationship between LPI and the average vertical GRF during the attenuation phase also coincided with a significant and strong association between LPI and the attenuation rate. We should keep in mind that the attenuation rate is a negative value, so a greater rate should align with a greater LPI. A greater rate of attenuation requires a greater peak vertical GRF, smaller vertical GRF at the end of attenuation, or both. Thus, an ideal approach for practitioners who prefer to focus on the attenuation rate during testing may be to target force curves with shorter attenuation times and smaller vertical GRF magnitudes at the end of the attenuation phase as opposed to larger peak vertical GRF values. These results highlight a potential goal to enhance athletes' ability to rapidly attenuate the

Table 2

Results for landing performance, momentum, height, and depth along with their associations with landing performance.*

	Landing performance index				Landing momentum (kg × m·s^{-1})				Landing height (m)				Landing depth (m)			
Subject	Mean	SD	*R*	*P*	Mean	*SD*	*R*	*p*	Mean	*SD*	*R*	*p*	Mean	*SD*	*R*	*p*
1	1.47	0.54	—	—	−232.67	7.76	−0.420	0.058	0.39	0.02	0.451	0.040	**−0.32**	**0.05**	**0.803**	**0.000**
2	1.26	0.36	—	—	**−255.44**	**24.85**	**−0.877**	**0.000**	**0.33**	**0.06**	**0.876**	**0.000**	−0.34	0.03	−0.124	0.563
3	1.23	0.34	—	—	−251.46	27.15	−0.386	0.063	0.39	0.08	0.368	0.077	−0.34	0.06	0.077	0.721
4	0.86	0.11	—	—	−249.85	6.69	−0.418	0.059	0.36	0.02	0.282	0.215	−0.40	0.04	0.178	0.440
5	1.57	0.21	—	—	−236.10	26.47	−0.072	0.824	0.28	0.06	0.051	0.875	−0.24	0.08	0.335	0.288
6	1.52	0.22	—	—	**−258.98**	**9.49**	**−0.756**	**0.019**	**0.40**	**0.03**	**0.714**	**0.031**	**−0.34**	**0.04**	**0.597**	**0.089**
7	1.89	0.23	—	—	−261.11	17.77	−0.423	0.040	0.41	0.06	0.442	0.031	−0.33	0.05	0.034	0.875
8	2.01	0.43	—	—	**−225.84**	**33.43**	**−0.820**	**0.001**	**0.41**	**0.12**	**0.839**	**0.001**	−0.28	0.07	−0.482	0.112
9	1.55	0.30	—	—	−265.12	17.36	−0.223	0.331	0.29	0.04	0.127	0.583	**−0.21**	**0.05**	**0.556**	**0.009**
10	1.42	0.27	—	—	−244.05	31.92	−0.459	0.036	0.35	0.07	0.499	0.021	−0.31	0.07	0.048	0.838
11	1.33	0.20	—	—	**−240.38**	**15.01**	**−0.760**	**0.001**	**0.38**	**0.04**	**0.795**	**0.000**	−0.37	0.03	0.042	0.881
12	2.05	0.33	—	—	**−272.78**	**14.30**	**−0.674**	**0.002**	**0.43**	**0.05**	**0.654**	**0.003**	−0.32	0.04	0.433	0.073
13	1.29	0.51	—	—	**−231.93**	**34.13**	**−0.648**	**0.001**	**0.30**	**0.08**	**0.649**	**0.001**	−0.26	0.05	0.030	0.891
14	1.75	0.54	—	—	**−255.33**	**17.53**	**−0.667**	**0.000**	**0.31**	**0.04**	**0.656**	**0.001**	**−0.23**	**0.04**	**0.686**	**0.000**
Group	1.51	0.33	—	—	−248.65	13.92	−0.184	0.529	0.36	0.05	0.335	0.242	−0.31	0.06	0.440	0.116

*Mean = average across trials (subject) or across subject averages (group); *SD* = ± one standard deviation across trials (subject) or across subject averages (group); *R* = correlation coefficient for the relationship with landing performance index; *p* = statistical probability for the relationship with landing performance index; bolded data = the relationship with landing performance index is statistically significant ($p < 0.05$) and of at least large magnitude ($R < 0.5$).

vertical GRF during landing with parallel considerations for the force magnitudes.

The approach we recommend above for practitioners seeking global improvements in the attenuation phase might seem odd considering that the relationship between LPI and the average vertical GRF during the control phase was very strong, suggesting that greater average force during that phase is likely to occur alongside greater LPI. However, this result should be taken into consideration with the very strong and significant relationship between LPI and the amortization vertical GRF, as a greater force value at the end of landing would increase the average vertical GRF during the control phase and account for smaller vertical GRF at the end of attenuation, presuming that the latter was not considerably small. These associations between LPI and the various average vertical GRF during the phases can help practitioners narrow in on where greater force application should be targeted during a landing. This is because the strong and significant relationship between LPI and the average vertical GRF during the entire landing alludes to more force output being ideal, but only when it occurs after the loading phase. By understanding these relationships between LPI and average vertical GRF during the various phases of landing, practitioners can develop training protocols to optimize force application and attenuation strategies, leading to improved athletic potential, physical development, or both.

Despite our acknowledgement in the introduction that musculoskeletal injury risk is difficult to link to vertical GRF data during landing, the logic behind greater risk and more rapidly occurring larger impact forces is reasonable. The peak vertical GRF during loading and the loading rate, metrics long-considered "higher risk," were not strongly associated

Table 3

Results for the landing phase durations along with their associations with landing performance.*†

	Landing time (s)				Loading time (s)				Attenuation time (s)				Control time (s)			
Subject	Mean	SD	*R*	*p*	Mean	*SD*	*R*	*p*	Mean	*SD*	*R*	*p*	Mean	*SD*	*R*	*p*
1	**0.28**	**0.07**	**−0.864**	**0.000**	0.06	0.01	−0.442	0.045	**0.22**	**0.07**	**−0.836**	**0.000**	0.07	0.06	−0.374	0.095
2	**0.27**	**0.05**	**−0.759**	**0.000**	0.06	0.01	−0.377	0.069	**0.21**	**0.05**	**−0.693**	**0.000**	0.15	0.08	−0.423	0.039
3	**0.34**	**0.12**	**−0.636**	**0.001**	0.06	0.01	−0.376	0.070	**0.28**	**0.13**	**−0.570**	**0.004**	0.14	0.14	−0.209	0.352
4	**0.42**	**0.05**	**−0.868**	**0.000**	0.06	0.01	−0.361	0.108	**0.36**	**0.05**	**−0.824**	**0.000**	**0.16**	**0.05**	**−0.606**	**0.004**
5	0.18	0.05	−0.444	0.148	0.07	0.01	−0.422	0.172	0.11	0.05	−0.334	0.289	0.06	0.05	−0.270	0.397
6	**0.27**	**0.03**	**−0.933**	**0.000**	0.06	0.01	−0.451	0.223	**0.21**	**0.03**	**−0.801**	**0.010**	**0.05**	**0.06**	**−0.587**	**0.097**
7	0.22	0.03	−0.377	0.069	0.06	0.02	0.457	0.025	0.16	0.04	−0.481	0.017	0.11	0.05	−0.272	0.198
8	0.20	0.03	0.171	0.594	**0.05**	**0.01**	**−0.547**	**0.066**	0.15	0.04	0.251	0.432	**0.07**	**0.06**	**0.559**	**0.074**
9	**0.19**	**0.04**	**−0.789**	**0.000**	0.06	0.01	−0.292	0.199	**0.14**	**0.04**	**−0.732**	**0.000**	0.05	0.04	0.219	0.341
10	0.25	0.05	−0.455	0.038	0.07	0.01	−0.226	0.325	0.18	0.05	−0.406	0.068	0.04	0.06	0.020	0.935
11	**0.28**	**0.03**	**−0.628**	**0.012**	0.07	0.01	−0.026	0.926	**0.21**	**0.03**	**−0.558**	**0.031**	0.05	0.04	−0.077	0.795
12	**0.21**	**0.03**	**−0.723**	**0.001**	**0.05**	**0.01**	**−0.753**	**0.000**	**0.16**	**0.02**	**−0.667**	**0.003**	0.11	0.04	−0.201	0.424
13	**0.25**	**0.07**	**−0.693**	**0.000**	0.08	0.01	0.102	0.635	**0.18**	**0.07**	**−0.687**	**0.000**	0.03	0.04	−0.470	0.027
14	**0.21**	**0.14**	**−0.744**	**0.000**	0.06	0.01	0.253	0.233	**0.14**	**0.15**	**−0.724**	**0.000**	**0.06**	**0.10**	**−0.584**	**0.003**
Group	**0.26**	**0.06**	**−0.803**	**0.001**	0.06	0.01	−0.497	0.070	**0.19**	**0.07**	**−0.759**	**0.002**	0.08	0.04	−0.275	0.341

*Mean = average across trials (subject) or across subject averages (group); *SD* = ± one standard deviation across trials (subject) or across subject averages (group); *R* = correlation coefficient for the relationship with landing performance index; *p* = statistical probability for the relationship with landing performance index.

†Bold values indicate the relationship with landing performance index is statistically significant ($p < 0.05$) and of at least large magnitude ($R < 0.5$).

Table 4

Results for the average vertical GRF magnitudes along with their associations with landing performance.*†

	Avg. landing GRF (xBW)				Avg. loading GRF (xBW)				Avg. attenuation GRF (xBW)				Avg. control GRF (xBW)			
Subject	Mean	SD	R	p	Mean	SD	R	p	Mean	SD	R	p	Mean	SD	R	p
1	**2.07**	**0.37**	**0.999**	**0.000**	**2.21**	**0.27**	**0.702**	**0.000**	**2.21**	**0.53**	**0.965**	**0.000**	**1.71**	**0.46**	**0.874**	**0.000**
2	**1.98**	**0.21**	**0.982**	**0.000**	1.80	0.20	0.394	0.057	**2.41**	**0.28**	**0.551**	**0.005**	**1.92**	**0.28**	**0.869**	**0.000**
3	**1.89**	**0.21**	**0.925**	**0.000**	2.24	0.29	0.268	0.205	2.29	0.54	0.177	0.407	**1.55**	**0.35**	**0.802**	**0.000**
4	**1.65**	**0.08**	**0.962**	**0.000**	2.08	0.14	−0.056	0.809	**1.75**	**0.09**	**0.571**	**0.007**	**1.39**	**0.11**	**0.848**	**0.000**
5	**2.35**	**0.22**	**0.746**	**0.005**	**2.20**	**0.30**	**0.552**	**0.063**	**2.66**	**0.31**	**0.789**	**0.002**	**2.11**	**0.21**	**0.536**	**0.137**
6	**2.08**	**0.14**	**0.986**	**0.000**	**2.09**	**0.16**	**0.740**	**0.023**	**2.18**	**0.11**	**0.838**	**0.005**	1.70	0.28	0.393	0.440
7	**2.34**	**0.15**	**0.808**	**0.000**	**1.96**	**0.19**	**0.581**	**0.003**	**2.78**	**0.39**	**0.726**	**0.000**	**2.47**	**0.38**	**0.722**	**0.000**
8	**2.43**	**0.17**	**0.734**	**0.007**	**2.25**	**0.32**	**0.945**	**0.000**	2.72	0.33	0.480	0.115	**2.41**	**0.15**	**0.616**	**0.141**
9	**2.32**	**0.25**	**0.950**	**0.000**	**2.20**	**0.22**	**0.692**	**0.001**	**2.77**	**0.54**	**0.812**	**0.000**	**2.11**	**0.21**	**0.855**	**0.000**
10	**2.10**	**0.19**	**0.840**	**0.000**	**1.99**	**0.18**	**0.598**	**0.004**	**2.25**	**0.34**	**0.634**	**0.002**	**1.94**	**0.28**	**0.723**	**0.012**
11	**1.98**	**0.11**	**0.946**	**0.000**	**1.84**	**0.14**	**0.749**	**0.001**	**2.13**	**0.15**	**0.691**	**0.004**	1.66	0.20	0.401	0.196
12	**2.41**	**0.19**	**0.953**	**0.000**	**2.07**	**0.22**	**0.781**	**0.000**	**2.87**	**0.26**	**0.775**	**0.000**	**2.44**	**0.19**	**0.798**	**0.000**
13	**2.06**	**0.34**	**0.930**	**0.000**	**2.16**	**0.29**	**0.789**	**0.000**	**2.15**	**0.45**	**0.736**	**0.000**	**1.53**	**0.41**	**0.789**	**0.002**
14	**2.43**	**0.40**	**0.973**	**0.000**	**2.20**	**0.24**	**0.885**	**0.000**	**2.93**	**0.66**	**0.860**	**0.000**	**2.27**	**0.51**	**0.891**	**0.000**
Group	**2.15**	**0.24**	**0.927**	**0.000**	2.09	0.15	0.205	0.482	**2.44**	**0.35**	**0.856**	**0.000**	**1.94**	**0.37**	**0.912**	**0.000**

*GRF = ground reaction force; BW = body mass; mean = average across trials (subject) or across subject averages (group); *SD* = ± one standard deviation across trials (subject) or across subject averages (group); *R* = correlation coefficient for the relationship with landing performance index; *p* = statistical probability for the relationship with landing performance index.

†Bold values indicate the relationship with landing performance index is statistically significant ($p < 0.05$) and of at least large magnitude ($R < 0.5$).

with LPI. This is an important result because it suggests that athletes with greater LPI are not likely exposed to greater GRF-related injury potential, although 50% of the athletes' results suggested that greater landing performance was associated with landing from higher heights and therefore greater mechanical demands. As mentioned in prior research (15), individuals who are better performers during landing use specific force attenuation and energy absorption strategies, and the current relationships provide some greater insight into where within the force curves practitioners might look to identify preferred strategies for both better performance and reduced force-related injury risk.

Specific, evidence-based training examples for the question why strength and conditioning professionals should assess landing performance is not yet available relative to landing performance as defined here. Still, we can extrapolate results from related work to show how a landing assessment can reveal positive intervention-based changes in landing performance. For example, when performers are instructed to use an external focus of attention (i.e., concentrate on pushing against the ground as rapidly as possible upon ground contact) during landing following a countermovement jump, landing height was shown to remain consistent, whereas landing time decreased in men and women, suggesting that a greater LPI was observed (25). Importantly, that same external focus coincided with smaller amounts of plantarflexion and knee flexion was observed at ground contact, suggesting greater preparatory flexion or dorsiflexion, which was followed by an increased loading time without increased peak vertical GRF (25). These results highlight the potential of

Table 5

Results for discrete vertical GRF magnitudes, loading rate, attenuation rate, and control rate along with their associations with landing performance.*†

	Peak loading GRF (xBW)				Loading rate (BW·s^{-1})				Attenuation rate (BW·s^{-1})				Control rate (BW·s^{-1})				Amortization GRF (xBW)			
Subject	Mean	SD	R	p	Mean	SD	R	p	Mean	SD	R	p	Mean	SD	R	p	Mean	SD	R	p
1	4.5	1.2	0.485	0.026	**73.3**	**28.8**	**0.626**	**0.002**	**−24.8**	**20.6**	**−0.805**	**0.000**	0.8	4.4	0.403	0.070	**1.7**	**0.5**	**0.846**	**0.000**
2	4.6	0.9	0.178	0.404	74.7	20.9	0.256	0.227	−73.1	38.8	−0.010	0.964	1.8	2.8	−0.089	0.681	**1.8**	**0.3**	**0.808**	**0.000**
3	6.1	1.6	−0.045	0.835	109.3	41.7	0.093	0.666	−59.7	54.3	−0.074	0.729	1.4	3.0	0.418	0.053	**1.5**	**0.4**	**0.654**	**0.001**
4	**4.3**	**0.7**	**0.573**	**0.007**	**74.5**	**18.1**	**0.581**	**0.006**	−15.9	4.5	−0.486	0.025	2.0	1.0	0.021	0.928	1.5	0.1	0.471	0.031
5	**4.4**	**0.8**	**0.609**	**0.036**	**59.5**	**16.6**	**0.672**	**0.017**	**−51.5**	**20.9**	**−0.586**	**0.045**	3.3	7.6	0.091	0.779	2.0	0.2	0.390	0.210
6	4.2	1.0	−0.024	0.950	78.3	35.4	0.095	0.808	−17.5	9.3	0.284	0.458	−0.6	4.2	−0.163	0.675	1.8	0.3	0.369	0.328
7	4.9	1.0	0.369	0.076	86.3	26.2	0.153	0.477	−74.2	47.5	−0.170	0.427	2.4	4.2	0.111	0.606	2.3	0.4	0.479	0.018
8	**5.2**	**1.2**	**0.870**	**0.000**	**102.5**	**37.2**	**0.834**	**0.001**	**−58.1**	**48.1**	**−0.758**	**0.004**	−2.7	8.2	0.485	0.131	2.1	0.3	0.244	0.444
9	**5.7**	**0.9**	**0.707**	**0.000**	**102.3**	**21.3**	**0.714**	**0.000**	**−71.8**	**42.4**	**−0.592**	**0.005**	2.0	7.2	0.317	0.161	**1.8**	**0.4**	**0.791**	**0.000**
10	3.5	0.5	0.369	0.100	51.5	12.3	0.364	0.105	−17.7	17.4	−0.299	0.187	−1.0	5.7	−0.185	0.448	**1.8**	**0.2**	**0.631**	**0.002**
11	3.7	0.5	0.192	0.492	53.1	13.8	0.135	0.630	−14.1	8.2	−0.051	0.857	0.3	1.8	−0.141	0.630	**1.7**	**0.2**	**0.506**	**0.054**
12	5.2	0.7	0.434	0.072	**94.8**	**20.8**	**0.579**	**0.012**	−79.7	30.1	0.011	0.965	2.0	4.4	0.257	0.303	**2.2**	**0.2**	**0.732**	**0.001**
13	**4.0**	**0.7**	**0.626**	**0.001**	**50.3**	**8.8**	**0.509**	**0.011**	−22.2	14.7	−0.489	0.015	**−3.8**	**7.7**	**−0.602**	**0.003**	**1.6**	**0.4**	**0.744**	**0.000**
14	**5.2**	**1.0**	**0.682**	**0.000**	86.4	27.1	−0.089	0.680	−60.4	34.4	−0.675	0.000	7.5	8.8	0.416	0.048	**2.3**	**0.6**	**0.854**	**0.000**
Group	4.7	0.7	0.332	0.246	78.3	19.7	0.400	0.155	**−45.8**	**25.5**	**−0.577**	**0.031**	1.1	2.7	0.113	0.700	**1.9**	**0.3**	**0.858**	**0.000**

*GRF = ground reaction force; BW = body mass; mean = average across trials (subject) or across subject averages (group); *SD* = ± one standard deviation across trials (subject) or across subject averages (group); *R* = correlation coefficient for the relationship with landing performance index; *p* = statistical probability for the relationship with landing performance index.

†Bold values indicate the relationship with landing performance index is statistically significant ($p < 0.05$) and of at least large magnitude ($R < 0.5$).

external foci of attention for increasing landing performance while also potentially reducing overuse injury risk.

Practical Applications

This article provides strength and conditioning practitioners who seek data-driven guidance for jump-landing assessments with important information as related to athlete performance tests. We demonstrate *how* to examine a force curve during a landing using appropriate phase deconstruction and *what* metrics should be focused on during a landing test before narrowing down a sport-specific, team-specific, or athlete-specific set of metrics. We also briefly touch on *why* strength and conditioning practitioners should assess landing performance because it can reveal an intervention's performance-related effects on an athlete's landing capacity. Although it is important to provide phase deconstruction methods that rely on sound biomechanical concepts, it is also important that the phases assessed provide actionable data. Because the countermovement jump includes a landing that is unavoidable, it is important that practitioners be provided with strategies to use *all* the data obtained during jump-landing tests. This work helps to provide such strategies, as we showed the loading, attenuation, and control phases each contain force-related and time-related metrics that can form a foundation upon which practitioners can identify critical strategies underlying landing performance. Although the metrics revealed herein to be associated with landing performance should not necessarily be applied to all athletes or even all basketball athletes, the results should be cautiously interpreted because of the exploratory nature of the analysis. Instead, practitioners can begin to conduct landing performance assessments, emphasizing some or all the following phase-specific metrics: landing time, attenuation time, average vertical GRF during landing, average vertical GRF during attenuation, average vertical GRF during control, vertical GRF attenuation rate, and the amortization GRF (i.e., GRF at zero velocity). Practitioners can then explore their own results to determine whether those metrics should be retained or replaced for their athletes or purposes. Overall, this article provides a framework for how to effectively use data from jump-landing tests to optimize training and improve landing performance.

Acknowledgments

The authors have no funding to declare. External resources did not support this project in the public, commercial, or not-for-profit sectors.

References

1. Barker LA, Harry JR, Mercer JA. Relationships between countermovement jump ground reaction forces and jump height, reactive strength index, and jump time. *J Strength Cond Res* 32: 248–254, 2018.
2. Bates BT. Single-subject methodology: An alternative approach. *Med Sci Sports Exerc* 28: 631–638, 1996.
3. Bates BT, Dufek JS, James CR, Harry JR, Eggleston JD. The influence of experimental design on the detection of performance differences. *Meas Phys Educ Exerc* 20: 200–207, 2016.
4. Bates BT, James CR, Dufek JS. Single subject analysis. In: *Innovative Analyses of Human Movement*. Stergiou N, ed. Champaign, IL: Human Kinetics, 2004. pp. 3–28.
5. Bishop C, Turner A, Jordan M, et al. A framework to guide practitioners for selecting metrics during the countermovement and drop jump tests. *Strength Cond J* 44: 95–103, 2022.
6. Bisseling RW, Hof AL, Bredeweg SW, Zwerver J, Mulder T. Relationship between landing strategy and patellar tendinopathy in volleyball. *Br J Sports Med* 41: e8, 2007.
7. Bressel E, Cronin J. The landing phase of a jump: Strategies to minimize injuries. *JOPERD* 76: 30–47, 2005.
8. Delextrat A, Badiella A, Saavedra V, et al. Match activity demands of elite Spanish female basketball players by playing position. *Int J Perform Anal Sport* 15: 687–703, 2015.
9. Devita P, Skelly WA. Effect of landing stiffness on joint kinetics and energetics in the lower extremity. *Med Sci Sports Exerc* 24: 108–115, 1992.
10. Dufek JS, Bates BT. The evaluation and prediction of impact forces during landings. *Med Sci Sports Exerc* 22: 370–377, 1990.
11. Dufek JS, Bates BT. Biomechanical factors associated with injury during landing in jump sports. *Sports Med* 12: 326–337, 1991.
12. Dufek JS, Bates BT, Stergiou N, James CR. Interactive effects between group and single-subject response patterns. *Hum Mov Sci* 14: 301–323, 1995.
13. Grozier CD, Cagle GK, Pantone L, et al. Effects of medial longitudinal arch flexibility on propulsion kinetics during drop vertical jumps. *J Biomech* 118: 110322, 2021.
14. Harry JR. MATLAB guide for analyzing countermovement strategies and performance over time. *Strength Cond J* 43: 44–53, 2021.
15. Harry JR, Barker LA, Eggleston JD, Dufek JS. Evaluating performance during maximum effort vertical jump landings. *J Appl Biomech* 34: 403–409, 2018.
16. Harry JR, Barker LA, James CR, Dufek JS. Performance differences among skilled soccer players of different playing positions during vertical jumping and landing. *J Strength Cond Res* 32: 304–312, 2018.
17. Harry JR, Barker LA, Mercer JA, Dufek JS. Vertical and horizontal impact force comparison during jump-landings with and without rotation in NCAA Division 1 male soccer players. *J Strength Cond Res* 31: 1780–1786, 2017.
18. Harry JR, Barker LA, Paquette MR. A joint power approach to identify countermovement jump phases using force platforms. *Med Scie Sports Exerc* 52: 993–1000, 2020.
19. Harry JR, Barker LA, Tinsley GM, et al. Relationships among countermovement vertical jump performance metrics, strategy variables, and inter-limb asymmetry in females. *Sports Biomech*: 1–9, 2021. In Press.
20. Harry JR, Blinch J, Barker LA, Krzyszkowski J, Chowning L. Low pass filter effects on metrics of countermovement vertical jump performance. *J Strength Cond Res* 36: 1459–1467, 2022.
21. Harry JR, Eggleston JD, Dufek JS, James CR. Single-subject analyses reveal altered performance and muscle activation during vertical jumping. *Biomechanics* 1: 15–28, 2021.
22. Harry JR, Freedman Silvernail J, Mercer JA, Dufek JS. Comparison of pre-contact joint kinematics and vertical impulse between vertical jump-landings and step-off landings from equal heights. *Hum Mov Sci* 56: 88–97, 2017.
23. Harry JR, Freedman Silvernail J, Mercer JA, Dufek JS. Bilateral comparison of vertical jump landings and step-off landings from equal heights. *J Strength Cond Res* 32: 1937–1947, 2018.
24. Harry JR, James CR, Dufek JS. Weighted vest effects on impact forces and joint work during vertical jump landings in men and women. *Hum Mov Sci* 63: 156–163, 2019.
25. Harry JR, Lanier R, Nunley B, Blinch J. Focus of attention effects on lower extremity biomechanics during vertical jump landings. *Hum Mov Sci* 68: 102521, 2019.
26. Harry JR, Paquette MR, Schilling BK, et al. Kinetic and electromyographic sub-phase characteristics with relation to countermovement vertical jump performance. *J Appl Biomech* 34: 291–297, 2018.
27. Hewett TE, Myer GD, Ford KR. Anterior cruciate ligament injuries in female athletes: Part 1, mechanisms and risk factors. *Am J Sports Med* 34: 299–311, 2006.
28. Hopkins WG, Marshall SW, Batterham AM, Hanin J. Progressive statistics for studies in sports medicine and exercise science. *Med Sci Sports Exerc* 41: 3–13, 2009.
29. James CR, Bates BT, Dufek JS. Classification and comparison of biomechanical response strategies for accommodating landing impact. *J Appl Biomech* 19: 106–118, 2003.
30. Kipp K, Kiely MT, Geiser CF. Reactive strength index modified is a valid measure of explosiveness in collegiate female volleyball players. *J Strength Cond Res* 30: 1341–1347, 2016.
31. Krzyszkowski J, Chowning LD, Harry JR. Phase-specific predictors of countermovement jump performance that distinguish good from poor jumpers. *J Strength Cond Res* 36: 3352–3358, 2022.

32. Lake JP, Mundy PD, Comfort P, et al. The effect of barbell load on vertical jump landing force-time characteristics. *J Strength Cond Rese* 35: 25–32, 2021.
33. Lees A. Methods of impact absorption when landing from a jump. *Eng Med* 10: 207–211, 1981.
34. Makaruk H, Czaplicki A, Sacewicz T, Sadowski J. The effects of single versus repeated plyometrics on landing biomechanics and jumping performance in men. *Biol Sport* 31: 9–14, 2014.
35. McMahon J, Suchomel TJ, Lake J, Comfort P. Understanding the key phases of the countermovement jump force-time curve. *Strength Cond J* 40: 96–106, 2018.
36. McNitt-Gray J. Kinematics and impulse characteristics of drop landing from three heights. *Int J Sports Biomech* 7: 201–224, 1991.
37. McNitt-Gray JL. Kinetics of the lower extremities during drop landings from three heights. *J Biomech* 26: 1037–1046, 1993.
38. McNitt-Gray JL, Hester DME, Mathiyakom W, Munkasy BA. Mechanical demand and multijoint control during landing depend on orientation of the body segments relative to the reaction force. *J Biomech* 34: 1471–1482, 2001.
39. Mizrahi J, Susak Z. Analysis of parameters affecting impact force attenuation during landing in human vertical free fall. *Eng Med* 11: 141–147, 1982.
40. Norcross MF, Lewek MD, Padua DA, et al. Lower extremity energy absorption and biomechanics during landing, Part II: Frontal-plane energy analyses and interplanar relationships. *J Athl Train* 58: 757–763, 2013.
41. Nordin AD, Dufek JS, James CR, Bates BT. Classifying performer strategies in drop landing activities. *J Sports Sci* 35: 1858–1863, 2017.
42. O'Keefe DJ. Colloquy: Should familywise alpha be adjusted? Against familywise alpha adjustment. *Hum Commun Res* 29: 431–447, 2003.
43. Reboussin DM, Morgan TM. Statistical considerations in the use and analysis of single-subject designs. *Med Sci Sports Exerc* 28: 639–644, 1996.
44. Rowley KM, Richards JG. Increasing plantarflexion angle during landing reduces vertical ground reaction forces, loading rates and the hip's contribution to support moment within participants. *J Sports Sci* 33: 1922–1932, 2015.
45. Schot PK, Dufek JS. Landing performance, Part I: Kinematic, kinetic, and neuromuscular aspects. *Med Exerc Nutr Health* 2: 69–83, 1993.
46. Self BP, Paine D. Ankle biomechanics during four landing techniques. *Med Sci Sports Exerc* 33: 1338–1344, 2001.
47. Taylor JB, James N, Mellalieu SD. Notational analysis of corner kicks in English premier league soccer. *Sci Football V*: 229–234, 2005.
48. Valls-Solé J, Rothwell JC, Goulart F, Cossu G, Munoz E. Patterned ballistic movements triggered by a startle in healthy humans. *J Physiol* 516: 931–938, 1999.
49. Zech A, Argubi-Wollesen A, Rahlf A-L. Minimalist, standard and no footwear on static and dynamic postural stability following jump landing. *Eur J Sport Sci* 15: 279–285, 2015.
50. Zhang SN, Bates BT, Dufek JS. Contributions of lower extremity joints to energy dissipation during landings. *Med Sci Sports Exerc* 32: 812–819, 2000.